

Sunil Kumar | Google Ph.D. Fellow

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Research Interest

Power Management & Performance Profiling: Specialized in maximizing HPC, ML, and LLM power efficiency (Perf/Watt) via DVFS, uncore frequency scaling, power capping, and thermal throttling. Experienced in microarchitectural profiling using perf and MSRs to classify compute-, memory-, and I/O-bound workloads for optimal P/C/T-state and power governor selection on modern SoCs. Developed a user-level tool enabling automated runtime power management for unseen multi-threaded workloads, significantly improving system throughput and energy efficiency across modern computing platforms.

Technical Skills

Programming Langs. (C/C++, Java, Python, Shell, Git) | **Parallel Runtimes** (OpenMP, POSIX Pthread, CUDA, MPI, PyTorch) | **Profiling Tools** (Linux-perf, LIKWID, Intel/AMD MSR PMCs) | **SW/HW power savings knobs** (ACPI CPU governor, DVFS, UFS, PCAP, DDCM, C-States, turbostats)

Education

Indraprastha Institute of Information Technology, Delhi July 2021 – Present
Ph.D. in Computer Science and Engineering CGPA: 8.69

Indraprastha Institute of Information Technology, Delhi August 2017 – April 2021
B.Tech. in Computer Science and Engineering CGPA: 7.0

Work Experience

Google Ph.D. Fellow @HiPeC Lab, IIIT Delhi July 2021 – Present
My thesis focuses on harnessing and advancing state-of-the-art runtime power-management techniques for exascale and AI-era computing systems, motivated by the rapidly growing power demands of modern data centers, cloud infrastructures, and supercomputers.

Research Assistant @HiPeC Lab, IIIT Delhi May 2021 – June 2021
Worked on a research project focused on building an energy-efficient MPI+OpenMP hybrid runtime for distributed systems and homogeneous multi-node clusters.

B.Tech. Thesis Researcher @IIIT-Delhi June 2020 – April 2021
In collaboration with Lawrence Berkeley National Laboratory (LBNL), USA
Built an energy-efficient parallel runtime in C/C++ for OpenMP applications, achieving 19.4% energy savings with a worst-case performance penalty of 3.6% on a 40-core Intel Xeon Haswell server.

Peer Reviewed Publications

Sunil Kumar and Vivek Kumar, "Reward-Driven Power Scheduling under Power Capping with DVFS-UFS Co-Optimization," in *under Review*, July 2026

Sunil Kumar, Vivek Kumar, and Sridutt Bhalachandra, "Power Scheduling for Maximizing Throughput and Fairness in Co-running Applications," in *ACM Transactions on Architecture and Code Optimization (TACO)*, May 2026. [\[Link\]](#)

Sunil Kumar and Vivek Kumar, "KarmaPM: Reward-Driven Power Manager," in *Proceedings of the European Conference on Parallel and Distributed Computing (Euro-Par'25)*, Dresden, Germany, August 2025. [\[Link\]](#)

Sunil Kumar, Akshat Gupta, Vivek Kumar, and Sridutt Bhalachandra, "Cuttlefish: Library for Achieving Energy Efficiency in Multicore Parallel Programs," in *Proceedings of the Supercomputing Conference (SC'21)*, St. Louis, MO, USA, November 2021. [\[Link\]](#)

Patents

Reward-based Bidirectional Power Scheduling on Multiprocessor Systems for Maximizing Throughput and Fairness

Filed with IIIT Delhi in *Indian Patent Office*. Based on the paper KarmaPM.
Filing date – October 07, 2025

Awards and Achievements

Euro-Par Travel Scholarship	2025
HiPC SRS Travel Support	2024
HiPC SRS Travel Support	2022
Google Ph.D. Fellowship (one of six awardees globally in the systems and networking domain) [Link]	2022
Best Artifact Reproducibility Finalist (one of five finalists for an SC conference paper)	2021
SIGHPC Travel Grant (one of seven awardees globally for an SC conference paper) [Link]	2021
Outstanding B.Tech Project Award	2021

Student Research/Doctoral Symposium Publications

Sunil Kumar, "Power Scheduling on Multicore Multiprocessor Systems for Maximizing Throughput and Fairness," in *PhD Symposium, Euro-Par'25*, Dresden, Germany, August 2025.

Sunil Kumar, Vivek Kumar, and Sridutt Bhalachandra, "RPM: Reward Power Manager for Power Distribution over a Cluster," in *Student Research Symposium, HiPC'24*. [\[Link\]](#)

Sunil Kumar, Vivek Kumar, and Sridutt Bhalachandra, "Energy Efficiency under Limited Power Budget," in *Student Research Symposium, HiPC'22*. [\[Link\]](#)

Teaching Experience

GPU Programming Specialization	<i>TAship Winter, Monsoon 2025</i>
Managed the system related issues occurred in CUDA programming and testing	
Operating Systems	<i>TAship Monsoon 2024</i>
Supported the course on kernel-level threading and memory management in C language	
Parallel Runtimes for Modern Processors	<i>TAship Winter 2024</i>
Conducted tutorial on GDB debugger and memory leakage detection using valgrind tool	
Operating Systems	<i>TAship Monsoon 2022</i>
Conducted weekly tutorials on system calls, GNU Makefile, Bash scripting, VM installations	
Foundations of Parallel Programming	<i>TAship Winter 2022</i>
Supported the course on developing runtimes for parallel programs and testing scalability on NUMA architecture	
Advanced Programming	<i>TAship Monsoon 2021</i>
Supported the course on OOPs concept in Java and android game developement using javafx library	

Invited Talks and Poster Presentations

Doctoral Symposium Lightning Talk	<i>Euro-Par, August 2025</i>
Research Meetup Talk featuring Prof. Vivek Sarkar, Georgia Tech, USA	<i>IIIT Delhi, July 2025</i>
SRS Lightning Talk	<i>HiPC, December 2024</i>
Research Meetup Talk featuring Dr. Suraj Kumar, INRIA, France	<i>IIIT Delhi, April 2024</i>
RIISE Poster Presentation	<i>IIIT Delhi, May 2023</i>
RIISE Talk	<i>IIIT Delhi, March 2022</i>
Invited Talk in Foundation of Parallel Programming Course	<i>IIIT Delhi, March 2022</i>
Invited Alumni Talk at Summer School	<i>IIIT Delhi, May 2021</i>
Guiding students on how to start research career in graduation	
Invited Talk in Foundation of Parallel Programming Course	<i>IIIT Delhi, March 2021</i>

Co-supervised B.Tech./M.Tech. Thesis Research Projects

Thermal-Aware Speed Control of Cooling Fans in Multicore Servers	<i>M.Tech., 2026</i>
Optimizing cooling costs by operating individual fans at variable speeds instead of a uniform speed	
Power-Aware Execution of LLM Inference on CPU Accelerators	<i>B.Tech., 2026</i>
Dynamically allocating power between compute and memory resources based on their requirements during the prefill and decode phases	
Energy-Efficient SIMD Phases in LLM Inference on CPU Accelerators	<i>B.Tech., 2025</i>
Mapping SIMD layers to specific instruction sets (SSE, AVX, AVX2, AVX-512, and AMX) based on their power and performance characteristics	
Analysis of Thermal Variations across Diverse HPC and ML Workloads	<i>B.Tech., 2025</i>
Evaluated thermal variations across diverse HPC and ML workloads on a quad-socket Intel Xeon 3rd-generation server with 144 cores	
Profiling Thermal Sensors on Motherboards Using IPMI Tools	<i>B.Tech., 2025</i>
Analyzed platform-level power consumption, including compute and cooling power, using off-chip thermal sensors and linux IPMI tools	
Maximizing Performance under Power Capping Using Bayesian Optimization	<i>B.Tech., 2024</i>
Developed an ML-based Bayesian optimization approach to maximize application performance under a fixed power budget	
Energy Profiling of SIMD Programs Using OpenBLAS and Intel MKL	<i>B.Tech., 2024</i>
Performed comparative power and performance analysis of SIMD programs using OpenBLAS and Intel MKL	
Energy and Performance Profiling of GPUs Using NVML and CUPTI Libraries	<i>B.Tech., 2022</i>
Evaluated GPU energy and performance profiling techniques on NVIDIA G100 and V100 GPUs	